

Design, Simulation, and Analysis of AM and FM Communication Systems

Due: Nov 25, 2024 (Monday) 11:59 pm

Project: Design and simulate both Amplitude Modulation (AM) and Frequency Modulation (FM) communication systems. The project will allow you to compare the performance, efficiency, and bandwidth requirements of AM and FM systems under various channel conditions.

A. Objective: This project aims to help you understand the core concepts of modulation and their applications in communication systems, providing a hands-on approach to comparing AM and FM systems under practical scenarios.

B. Key Components:

1. **System Design:**

- **AM System:**
 - Design the basic architecture of an AM transmitter and receiver.
 - Implement the modulation of a carrier signal using a baseband message (e.g., an audio signal).
 - Include components like the modulator, demodulator, and low-pass filter.
- **FM System:**
 - Design the architecture of an FM transmitter and receiver.
 - Modulate the carrier frequency based on the input signal's amplitude.
 - Implement demodulation to retrieve the original message signal.

1. **Simulation Tools:**

- Use MATLAB, or any preferred simulation tool to implement the AM and FM systems.
- Visualize the modulation and demodulation process using time-domain waveforms and frequency-domain spectrums.

2. **Performance Metrics:**

- **Signal-to-Noise Ratio (SNR):** Analyze how noise affects AM and FM signals in the communication channel.
- **Bandwidth Efficiency:** Measure the bandwidth used by AM and FM systems and compare their efficiency.
- **Power Efficiency:** Compare the power consumption of AM and FM transmitters.

- **Distortion and Interference:** Evaluate the sensitivity of each modulation type to interference and channel noise.
3. **Comparison and Analysis:**
- **Noise Immunity:** Compare how AM and FM signals perform in noisy environments. Demonstrate that FM has better noise resistance than AM.
 - **Bandwidth Requirements:** Analyze the bandwidth requirements of both systems. AM generally requires less bandwidth, while FM uses more but offers better signal quality.
 - **Audio Quality:** Simulate transmitting an audio signal and compare the quality of AM and FM demodulation.
4. **Channel Simulation:**
- Introduce channel effects like additive white Gaussian noise (AWGN).
 - Study the impact of these effects on the transmitted AM and FM signals.
5. **Results and Visualization:**
- Plot waveforms showing the modulation and demodulation processes for both AM and FM.
 - Use frequency spectrum plots to show the bandwidth occupied by each system.
 - Compare the effects of noise on the AM and FM signals in different SNR conditions.

C. Learning Outcomes:

1. Understand the principles behind AM and FM modulation techniques.
2. Gain hands-on experience in designing and simulating modulation systems.
3. Compare the advantages and disadvantages of AM and FM in terms of bandwidth usage and noise resistance.
4. Learn how to simulate communication channels and analyze system performance under various conditions.

D. **Additional Features (Optional):**

- Implement more advanced modulation techniques such as
 - **AM: DSB-SC and SSB (LSB or USB)** for further comparison.
 - **FM: NBFM (Narrowband FM)** for further comparison.
- Include error correction or filtering techniques to improve system performance under noisy conditions.